

Methodological Note: Facility Readiness, Accessibility, and a Governed Spatial Database

Question answered

Given a facility personnel-adequacy score, where are populations underserved — and is that because no facility is physically near them, or because a nearby facility exists but is inadequately staffed? These require different interventions (construction vs. staffing vs. assessment), so the pipeline keeps them distinct throughout rather than collapsing to a single access number.

Data reconciliation (full detail in outputs/logs/)

The facility register, personnel scoring file, senatorial-district crosswalk, and boundary layer did not agree with one another out of the box. Nothing was dropped without being counted: 18 systematically duplicated facility records removed; coordinates parsed from 3 mixed formats, 24 missing entirely (23 recovered from the score file's independent geometry, 1 genuinely unlocatable); 9 facilities had register lon/lat swapped and 7 had coordinates hundreds of km outside the country, both corrected using the score file's cross-validated geometry; 9 placeholder score rows excluded; 106 facilities never assessed (kept null, not zero); the crosswalk had 2 duplicate rows and 1 genuine conflict, resolved against the boundary layer.

Database, CRS, and access measure

Database: DuckDB with the spatial extension — embedded, enforces declared primary/foreign keys, supports RTree indexes, avoiding both a PostGIS server and GeoPackage's lack of enforced constraints. Facilities assigned to wards by point-in-polygon (nearest-ward fallback at edges), cross-checked against the register's stated ward name: 0 mismatches out of 1,327.

CRS: EPSG:32632 (UTM 32N). The area straddles the 6°E zone boundary; tested against a custom Transverse Mercator using true WGS84 geodesic distance as ground truth (300 random pairs): UTM 32N mean error 0.027% (max 0.091%), custom TM mean 0.018% (max 0.085%). Immaterial at the km-scale thresholds used here, so the standard, independently-verifiable zone was kept.

Access measure: straight-line 5 km catchment. Network distance was evaluated and rejected as primary: median facility-to-road distance is 7.3 km — greater than the catchment threshold itself — so it would mostly reflect gaps in a sparse 213-feature digitization rather than real accessibility. The 5 km threshold is externally grounded (WHO-consistent primary-care catchment guidance) and consistent with this dataset's own facility spacing (median nearest-neighbour distance: 5.8 km). Population is assumed uniformly distributed within each ward (no gridded surface supplied) — a stated simplification, not a claim of precision. Adequacy follows minimum_staffing_norms.adequacy_rule exactly: adequately staffed when actual counts meet or exceed the minimum for every cadre with a non-zero minimum for its facility type.

Ward classification (620 wards, 22,936,947 people)

Category	Wards	Meaning
No facility within 5km	7	Genuinely absent — construction problem
Facility nearby, confirmed inadequate	120	Assessed and understaffed — staffing problem
Facility nearby, assessment gap	12	Never assessed — data-collection problem
Facility nearby, mixed adequacy	363	Partial coverage from a mix of facilities
Facility nearby, adequately staffed	118	Genuinely well served

Thresholds (adequate-coverage share $\leq 10\%$ / $\geq 80\%$ of a ward's facility-accessible population) were set from the data's own distribution — a real concentration at both ends among the 613 wards with any facility access — not asserted round numbers.

Headline finding: only 18.4% of the population is within 5 km of an adequately staffed facility, against 39.8% within 5 km of *any* facility — the staffing gap, not the physical access gap, is the larger problem in this dataset. Agbkeko North senatorial district is the clearest priority: 62.5% of its population sits in confirmed-inadequate-coverage wards, the highest of any district, with essentially no adequately-staffed reach at all.

Known limitations

- Uniform within-ward population distribution assumed; a gridded population surface would sharpen the population-weighted figures.
- Straight-line catchment ignores terrain, rivers, and actual travel time; the supplied road network was too sparse to substitute a credible network-distance measure.
- Single cross-sectional snapshot; no trend or seasonal variation (e.g. wet-season road access) is captured.
- 106 facilities' staffing adequacy is unknown, not zero; wards dominated by these are flagged as an assessment gap, not assumed inadequate — but the true state of care there is genuinely unknown from this data.
- The 5km / 10% / 80% thresholds are defensible and data-grounded but remain choices; underlying continuous measures are retained in ward_classification.csv for anyone who wants to re-cut them.